

Curry–Howard Correspondences for Algebraic Structures in CAL

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1. Vision

Core Question:

When does an algebra admit a computational interpretation?

The Curry–Howard correspondence connects:

- Proofs $\leftrightarrow$ Programs
- Propositions $\leftrightarrow$ Types
- Normalization $\leftrightarrow$ Computation

It is well understood for:

- Intuitionistic logic
- Linear logic
- Classical logic (not so much)

Goal: Extend Curry–Howard systematically to algebraic structures in the CAL database.

2. Why CAL?

Peter Jipsen's *Categories, Algebras and Logics (CAL)* database catalogues hundreds of algebraic varieties:

- Residuated lattices
- BCK/BCI-algebras
- MV-algebras
- Bilattices
- Boolean algebras with operators
- Many more

Each comes with:

- Algebraic axioms
- Logical counterparts
- Structural properties

Missing piece: A systematic proof-as-programs interpretation.

3. Research Program

For selected families in CAL or of the EPS (Encyclopedia of Proof Systems):

- 1 Define appropriate proof systems
- 2 Construct typed term calculi
- 3 Develop categorical semantics
- 4 Prove normalization and soundness

Target families may include:

- Residuated structures (substructural logics)
- Implicational systems (BCK/BCI)
- Many-valued logics (MV-algebras)?
- Paraconsistent structures (bilattices)?

Deliverable: A taxonomy linking

Algebras $\leftrightarrow$ Logics $\leftrightarrow$ Calculi $\leftrightarrow$ Categories.

4. Technical Strategy

Algebraic Logic Tools

- Algebraizability
- Leibniz operator
- Varieties and quasivarieties

Proof Theory

- Sequent calculi
- Natural deduction
- Structural rule analysis

Categorical Semantics

- Monoidal and residuated categories
- Hyperdoctrines & doctrines
- Enriched and indexed categories
- fibrations

Key methodological principle:

Let algebra determine computation.

5. Expected Outcomes

- New Curry–Howard correspondences beyond classical cases
- Structural criteria for when CH exists
- Prototype calculi and implementations
- Publications bridging algebraic logic and type theory

Long-term vision:

A general theory of computational interpretations of algebraic varieties.

Algebra $\Rightarrow$ Logic $\Rightarrow$ Computation